

CMP-142 Object Oriented Programming

Lab-09

Topics:

- Exception Handling

Instructions!

- This is a **graded lab**, you are strictly **NOT** allowed to discuss your solutions with your fellow colleagues, even not allowed to ask how he/she is doing, it may result in negative marking. You can **ONLY** discuss with your TAs or instructor.
- Indent your code properly.
- Use meaningful variable and function names. **Use the camelCase notation.**
- Save your work frequently. Make a habit of pressing CTRL+S after every line of code you write.
- **Do NOT use any global or static variable(s).** However, global named constants may be used.
- Make sure that there are **NO dangling pointers or memory leaks** in your programs.
- Make sure that all functions have been implemented and can be called in the main function. Otherwise, no marks will be awarded for that function.
- ***Prototypes of some functions are omitted. You are supposed to complete them on your own. Moreover, there might be a case where some functions are ignored altogether while giving structure of class. You have to write and complete them.***

[Task 1] Inventory Management with Exception Handling [50 Marks]

Objective:

Create an Inventory Management System that tracks products in a warehouse, manages stock levels, handles multiple types of operations (add, remove, check availability), and ensures proper exception handling in case of errors such as invalid stock operations, negative quantities, or missing items.

Task Description:

1. Product Class:

Create a Product class that contains the following attributes and functions:

```
productID: A unique identifier for each product (integer).  
name: The product name (string).  
quantity: The current quantity in stock (integer).  
price: The price per unit of the product (float).
```

```
Add stock: Add a specified quantity to the stock.  
Remove stock: Remove a specified quantity from the stock.  
Check availability: Check if the requested quantity is available in  
stock.  
Display product info: Show the product's details (ID, name, quantity,  
price)
```

2. Inventory Class:

Create an Inventory class to manage multiple products. The Inventory class should include the following attributes and functions:

```
Product** products;    //Array of pointers to Product objects  
size;                  // Current number of products in inventory  
capacity;              // Maximum capacity of the inventory
```

Add product: Add a new product to the inventory.

Remove product: Remove a product by productID.

Update product: Update the details of an existing product (e.g., name, price).

Update Stock: Update stock of a product

Check product availability: Check if a particular product exists and whether enough stock is available.

View product: View the details of a specific product.

View all products: Display all products in the inventory.

Save To File: Save inventory data to file

Load From File: Load inventory data from file

Exception Handling:

Negative Quantity: Throw an exception if a negative quantity is passed when adding or removing stock.

Out of Stock: If trying to remove more stock than is available, throw an exception

Invalid Product: When trying to access or modify a product that doesn't exist in the inventory, throw an exception.

Invalid Input: If the user enters invalid data (e.g., non-numeric input for quantity or price), catch and handle the exception gracefully.

File Errors: If loading/saving the inventory to/from a file, handle file opening errors (e.g., file not found or permissions issues).

User Interface:

The program should run in a loop and display the following options to the user:

1. Add a product.
2. Remove a product.
3. Add stock to an existing product.
4. Remove stock from an existing product.
5. Check product availability.
6. View product details.
7. View all products.
8. Save and exit.

Output:

```
Welcome to the Inventory Management System
1. Add a product
2. Remove a product
3. Add stock to an existing product
4. Remove stock from an existing product
5. Check product availability
6. View product details
7. View all products
8. Save and exit
Choose an option: 3
Enter product ID: 101
Enter quantity to add: 50

Product added successfully! Updated stock of Product ID 101: 50
units.

Choose an option: 4
Enter product ID: 101
Enter quantity to remove: 20

Product removed successfully! Updated stock of Product ID 101: 30
units.

Choose an option: 7
Product ID: 101 | Name: Widget | Quantity: 30 | Price: 9.99

Choose an option: 8
```

Saving data to file... Successfully saved inventory to inventory.txt.
Goodbye!